

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of BSc Physic Examination April-May 2018

PD150 Electricity and Magnetism

Date: 01-05-2018 Day: Tuesday Time: 10.00AM To 10:30AM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 20

1. Two electrically neutral materials are rubbed together. One acquires a net positive charge. The other must have:
(a) lost electrons
(b) gained electrons
(c) lost protons
(d) gained protons
2. What is the magnitude of the charge on a body that has an excess of 20 electrons?
(a) $3.2 \times 10^{-18} \text{ C}$
(b) $1.6 \times 10^{-18} \text{ C}$
(c) $3.2 \times 10^{-19} \text{ C}$
(d) $2.4 \times 10^{-19} \text{ C}$
3. What is the potential difference between point A and point B if 10 J of work is required to move a charge of 4C from one point to the other?
(a) 0.400 V
(b) 2.50 V
(c) 14.0 V
(d) 40.0 V
4. Two point charges, A and B, are separated by 10.0 m. If the distance between them is reduced to 5.00 m, the force exerted on each:

- (a) decreases to one-half its original value
 - (b) increases to twice its original value
 - (c) decreases to one-quarter of its original value
 - (d) increases to four times its original value
5. If three unequal capacitors, initially uncharged, are connected in series across a battery, which of the following statements is true?
- (a) The equivalent capacitance is greater than any of the individual capacitances
 - (b) The largest voltage appears across the capacitor with the smallest capacitance
 - (c) The largest voltage appears across the capacitor with the largest capacitance
 - (d) The capacitor with the largest capacitance has the greatest charge
6. An electronics technician wishes to construct a parallel-plate capacitor using titanium dioxide ($k=1 \times 10^2$) as the dielectric. If the cross-sectional area of the plates is 1 cm^2 , what is the capacitance if the rutile thickness is 1.00 mm ?
- (a) 88.5 pF
 - (b) 177.0 pF
 - (c) $8.85 \text{ }\mu\text{F}$
 - (d) $100.0 \text{ }\mu\text{F}$
7. A rectangular coil of wire consisting of ten loops, each with length 0.20 m and width 0.30 m , lies in the xy plane. If the coil carries a current of 2 A , what is the torque exerted by a magnetic field of magnitude 0.010 T directed at an angle of 30.0° with respect to the positive z -axis?
- (a) $1.2 \times 10^{-2} \text{ Nm}$
 - (b) $2.4 \times 10^{-2} \text{ Nm}$
 - (c) $6.0 \times 10^{-3} \text{ Nm}$
 - (d) $4.0 \times 10^{-3} \text{ Nm}$
8. A proton enters a constant magnetic field of magnitude 0.050 T and traverses a semicircle of radius 1.0 mm before leaving the field. What is the proton's speed?
- (a) $1.6 \times 10^3 \text{ m/s}$
 - (b) $2.3 \times 10^3 \text{ m/s}$
 - (c) $2.8 \times 10^4 \text{ m/s}$
 - (d) $4.8 \times 10^3 \text{ m/s}$
9. What is the magnitude of the magnetic field at the core of a 120-turn solenoid of length 0.50 m carrying a current of 2.0 A ?
- (a) $2.4 \times 10^{-4} \text{ T}$
 - (b) $4.8 \times 10^{-5} \text{ T}$
 - (c) $1.2 \times 10^{-6} \text{ T}$
 - (d) $6.0 \times 10^{-4} \text{ T}$
10. At a certain instant, a proton is moving in the positive x -direction through a magnetic field in the negative z -direction. What is the direction of the magnetic force exerted on the proton?
- (a) positive z -direction
 - (b) negative z -direction
 - (c) positive y -direction
 - (d) negative y -direction
11. A flat coil of wire is placed in a uniform magnetic field that is in the y -direction. The magnetic flux through the coil is a maximum if the plane of the coil is where?
- (a) in the xy -plane
 - (b) in the yz -plane
 - (c) in the xz -plane
 - (d) in any orientation, because it is a constant
12. The current in a 5.00-H inductor decreases uniformly at a rate of 2.00 A/s . What is the voltage drop across the inductor?

- (a) 2.50 V
 - (b) 5.00 V
 - (c) 0.400 V
 - (d) 10.0 V
13. What happens to the amplitude of the induced emf when the rate of rotation of a generator coil is doubled?
- (a) It becomes four times larger
 - (b) It becomes two times larger
 - (c) It is unchanged
 - (d) It becomes one-half as large
14. A series RLC circuit contains a resistor of 20 Ω , a capacitor of 0.75 μF , and an inductor of 120 mH. If a sinusoidally varying rms voltage of 120 V is applied across this combination of elements, what is the rms current in the circuit when operating at its resonance frequency?
- (a) 2.4 A
 - (b) 6.0 A
 - (c) 10 A
 - (d) 17 A
15. A typical microwave oven operates at a frequency of 2.45 GHz. What is the wavelength associated with the electromagnetic waves in the oven?
- (a) 8.20 m
 - (b) 12.2 cm
 - (c) 1.2×10^8 m
 - (d) 8.2×10^{-9} m
16. A resistor, capacitor, and inductor are connected in series across an AC generator. Which of the following statements is true?
- (a) The instantaneous voltage across the capacitor lags the current by 90°
 - (b) The instantaneous voltage across the inductor leads the current by 90°
 - (c) The instantaneous voltage across the resistor is in phase with the current
 - (d) All the above
17. What is the impedance of a series RLC circuit at resonance?
- (a) X_L
 - (b) X_C
 - (c) R
 - (d) $X_L - X_C$
18. Susceptibility is positive and large for:
- (a) paramagnetic
 - (b) diamagnetic
 - (c) ferromagnetic
 - (d) both (a) and (c)
19. Ballistic galvanometer is used:
- (a) to measure transient current or voltage of short duration
 - (b) to measure electromagnetic induction
 - (c) to measure the magnetic field
 - (d) to measure eddy current
20. The energy of an electron (or proton) gains or losses by moving through a potential difference of 1 V is:
- (a) $1.6 \times 10^{-19} \text{ J}$
 - (b) $1.6 \times 10^{19} \text{ J}$
 - (c) $1.6 \times 10^{-19} \text{ V}$
 - (d) $1.6 \times 10^{-19} \text{ C}$

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II Semester of BSc Physic Examination April-May 2018

PD150 Electricity and Magnetism

Date: 01-05-2018 Day: Tuesday Time: 10:30AM To 01:00PM Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

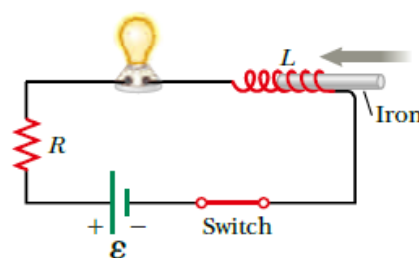
SECTION – I

Q – II Answer the following questions as directed

20

1. (i) What is the electric potential energy of two electrons separated by 2 nm? 2
(ii) If the separation increases, does the potential energy increase or decrease?
 2. Define (i) Superposition theorem and (ii) Reciprocity theorem associated with the network analysis. 2
 3. In the given figure, points A, B, and C are at various distances from a given point charge. At which point the electric field strength is highest. Give reasons. 2
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4. A parallel-plate capacitor is disconnected from a battery, and the plates are pulled a small distance farther apart. Do the following quantities increase, decrease, or stay the same? (i) C (ii) Q (iii) E between the plates (iv) energy stored in the capacitor 2
 5. (i) In an RC circuit, the capacitor begins to discharge. During the discharge, in the region of space between the plates of the capacitor, there is displacement current but no conduction current. State: True /False 2
(ii) A charged particle moves with velocity v in a magnetic field B . The magnetic force on the particle is maximum when v is parallel to B State: True /False
 6. Write Maxwell's equations of electromagnetism. 2
 7. Draw hysteresis loop ferromagnetic materials and explain retentivity and coercivity. 2
 8. (i) As a charged particle moves freely in a circular path in the presence of a constant magnetic field applied perpendicular to the particle's velocity, the particle's kinetic energy (a) remains constant, (b) increases, or (c) decreases. 2
(ii) Which of the following actions would double the magnitude of the magnetic force per unit length between two parallel current-carrying wires? Choose all correct answers. (a) Double one of the currents. (b) Double the distance between them. (c) Reduce the distance between them by half

9. The switch in the circuit shown in Figure is closed, and the lightbulb glows steadily. The inductor is a simple air-core solenoid. An iron rod is inserted into the interior of the solenoid, increasing the magnitude of the magnetic field in the solenoid. As the rod is inserted, the brightness of the lightbulb (a) increases, (b) decreases, or (c) remains the same.



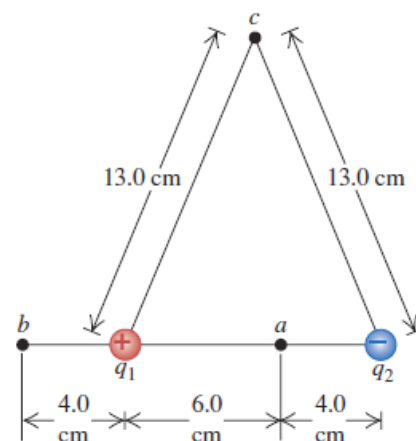
Expalin.

10. (i) If the resistance in an RLC circuit remains the same, but the capacitance and inductance are each doubled, how will the resonance frequency change?
 (ii) True or False: In an RLC circuit, the impedance must always be greater than or equal to the resistance

SECTION – II

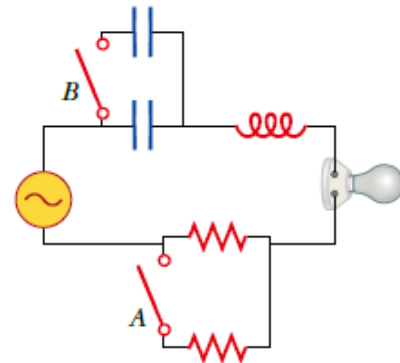
Q-III Answer the following questions as directed

1. Two point charges, $Q_1 = +25\text{nC}$ and $Q_2 = -75\text{nC}$, are separated by a distance $r = 3\text{cm}$. Find the magnitude and direction of the electric force (a) that Q_1 exerts on Q_2 and (b) Q_2 that exerts on Q_1 .
 2. Two charges, one of $2.5\mu\text{C}$ and the other of $-3.5\mu\text{C}$ are placed on the $-x$ -axis, one at the origin and the other at $x = 0.6\text{m}$. Find the position on the x -axis where the net force on a small charge $+q$ would be zero.
 3. A $6.2\mu\text{C}$ point charge is at the center of a cube with sides of length 0.5m . (a) What is the electric flux through one of the six faces of the cube? (b) How would your answer to part (a) change if the sides were 0.250 m long? Explain.
 4. An electric dipole consists of point charges $q_1 = +12\text{nC}$ and $q_2 = -12\text{nC}$ placed 10cm apart. Compute the electric potentials at points a, b and c.



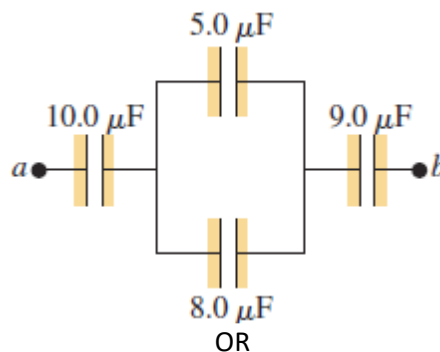
5. A parallel plate air capacitor is to store charge of magnitude 240 pC on each plate when the potential difference between the plates is 42V .
 (a) If the area of each plate is 6.8cm^2 , what is the separation between the plates?
 (b) If the separation between the two plates is double the value calculated in part (a) what potential difference is required for the capacitor to store charge of magnitude 240pC on each plate?

6. (i) If switch A is closed in Figure, what happens to the impedance of the circuit? (a) It increases (b) It decreases (c) It doesn't change
 (ii) Suppose $X_L > X_C$ in Figure. If switch A is closed, what happens to the phase angle? (a) It increases (b) It decreases (c) It doesn't change
 (iii) Suppose $X_L > X_C$ in Figure. If switch A is left open and switch B is closed, what happens to the phase angle? (a) It increases. (b) It decreases. (c) It doesn't change.

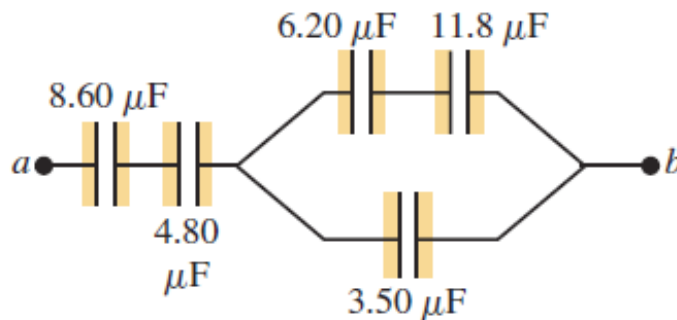


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7. Figure shows a system of four capacitors, where the potential difference across ab is 50 V.
 (a) Find the equivalent capacitance of this system between a and b
 (b) How much charge is stored by this combination of capacitors?
 (c) How much charge is stored in each of the $10\mu\text{F}$ and $9\mu\text{F}$ capacitors?
 (d) What is the total energy stored?



For the capacitor network shown in Figure, the potential difference across ab is 12.0 V. Find (a) the total energy stored in this network and (b) the energy stored in the $4.8\mu\text{F}$ capacitor.

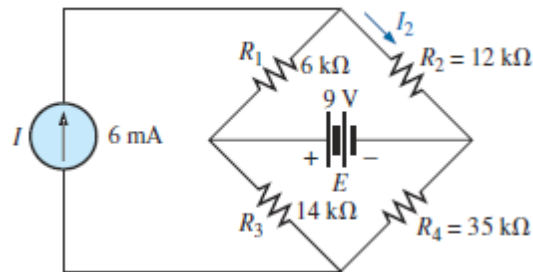


8. A particle with a charge of $-1.24 \times 10^{-8} \text{C}$ is moving with instantaneous velocity $\vec{v} = (4.19 \times 10^4 \text{m/s})\hat{i} + (-3.85 \times 10^4 \text{m/s})\hat{j}$. What is the force on this particle by a magnetic field (a) $\vec{B} = (1.4 \text{T})\hat{i}$ and (b) $\vec{B} = (1.4 \text{T})\hat{k}$ 4

OR

A coil 4 cm in radius, containing 500 turns, is placed in a uniform magnetic field that varies with time according to $B = (0.0120 \text{ T/s})t + (3 \times 10^{-5} \text{ T/s}^4)t^4$. The coil is connected to a 600Ω resistor, and its plane is perpendicular to the magnetic field. You can ignore the resistance of the coil. (a) Find the magnitude of the induced emf in the coil as a function of time. (b) What is the current in the resistor at time $t = 5 \text{s}$?

9. Using the principle of superposition, find the current I_2 through the $12 \text{k}\Omega$ resistor in the figure. 4



OR

Find the Thevenin equivalent circuit for the network external to the resistor R_4 in Figure.

